Contents

Chapter 1 Groups and homomorphisms

Exercise 1.1

Let N be a subgroup of G and $g \in G$. Then $g^{-1}Ng = \{g^{-1}ng \mid n \in N\} = \{g^{-1}gn \mid n \in N\} = \{n \mid n \in N\} = N$ because G is abelian. So any subgroup of G is normal. Since G is simple, the only subgroups of G are $\{1\}$ and G. Also, using the fact that G is simple we can assume that $G \neq \{1\}$. Choose $h \in G \setminus \{1\}$. Since $\langle h \rangle$ is a subgroup of G, $\langle h \rangle = G$ or $\langle h \rangle = \{1\}$. If $\langle h \rangle = \{1\}$, then $h = 1$, a contradiction. Hence, $\langle h \rangle = G$, so G is a cyclic group. Let $|G| = n$. Assume for contradiction that n is not prime, so $n = ab$ for some $1 < a, b < n$. Since G is cyclic and a is a factor of n, there exists $k \in G$ of order a. Since $|k| > 1$, $k \neq 1$. By the above argument, we have $\langle k \rangle = G$. Therefore, $a = |k| = |\langle k \rangle| = |G| = n$, a contradiction. Hence, G is cyclic of prime order. $\square$

Exercise 1.2

Assume $H \neq \{1\}$. Let $g_1, g_2 \in G$. Suppose $g_1\vartheta = g_2\vartheta$. Then $(g_1\vartheta)(g_2\vartheta)^{-1} = 1$ and thus $(g_1g_2^{-1})\vartheta = 1$. Hence $g_1g_2^{-1} \in \mathrm{Ker}(\vartheta)$. Since $\mathrm{Ker}(\vartheta) \lhd G$, we have $\mathrm{Ker}(\vartheta) = \{1\}$ or $\mathrm{Ker}(\vartheta) = G$. If $\mathrm{Ker}(\vartheta) = G$, then $g\vartheta = 1$ for all $g \in G$. Since ϑ is surjective, $H = \mathrm{Im}(\vartheta) = \{g\vartheta \mid g \in G\} = \{1\}$, a contradiction. Therefore, $\mathrm{Ker}(\vartheta) = \{1\}$. Hence, we have $g_1g_2^{-1} = 1$, which implies $g_1 = g_2$. So ϑ is injective and thus is bijective. $\square$

Chapter 2 Vector spaces and linear transformations

Exercise 2.1

Let $w, w' \in W$ and $\lambda \in F$. Since ϑ is invertible, ϑ is surjective, so there exist $v, v' \in V$ such that $v\vartheta = w$ and $v'\vartheta = w'$. Then we have $v = w\vartheta^{-1}$ and $v' = w'\vartheta^{-1}$. Note that

$$\begin{aligned}(w + w')\vartheta^{-1} &= (v\vartheta + v'\vartheta)\vartheta^{-1}\\ &= ((v + v')\vartheta)\vartheta^{-1}\\ &= v + v'\\ &= w\vartheta^{-1} + w'\vartheta^{-1}.\end{aligned}$$

Also, note that

$$\begin{aligned}(\lambda w)\vartheta^{-1} &= (\lambda(v\vartheta))\vartheta^{-1}\\ &= ((\lambda v)\vartheta)\vartheta^{-1}\\ &= \lambda v\\ &= \lambda(w\vartheta^{-1}).\end{aligned}$$

Therefore, ϑ^{-1} is a linear transformation. $\square$

Exercise 2.3

Suppose $V = U \oplus W$. Clearly, $V = U + W$. Let $v \in U \cap W \subseteq V$. Then there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. So $u + w \in U$ and $u + w \in W$. By the uniqueness of u and w, we have $u+w=u$ and $u+w=w$, so $u=w=0$, and thus $v=0$. Hence, $U \cap W = \{0\}$.

Now suppose $V = U + W$ and $U \cap W = \{0\}$. We want to prove that for any $v \in V$ there exist unique $u \in U$ and $w \in W$ such that $v = u+w$. Let $v \in V$. Suppose $u_1, u_2 \in U$ and $w_1, w_2 \in W$ satisfy $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. Since U and W are subspaces, they are closed under vector addition and additive inverses, so $u_1 - u_2 \in U$ and $w_2 - w_1 \in W$, so $u_1 - u_2 = w_2 - w_1 \in U \cap W$. Hence, $u_1 - u_2 = w_2 - w_1 = 0$, and thus $u_1 = u_2$ and $w_1 = w_2$, proving the uniqueness of u and w.

Therefore, $V = U \oplus W$ if and only if $V = U + W$ and $U \cap W = \{0\}$. $\square$

Exercise 2.4

Suppose $V = U \oplus W$. Let $v \in V$. Then there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. Since $u_1, \ldots, u_r$ is a basis of U, there exist unique $\lambda_1, \ldots, \lambda_r \in F$ such that $u = \lambda_1 u_1 + \ldots + \lambda_r u_r$. Since $w_1, \ldots, w_s$ is a basis of W, there exist unique $\mu_1, \ldots, \mu_s \in F$ such that $w = \mu_1 w_1 + \ldots + \mu_s w_s$. Therefore, there exist unique $\lambda_1, \ldots, \lambda_r, \mu_1, \ldots, \mu_s \in F$ such that $v = \lambda_1 u_1 + \ldots + \lambda_r u_r + \mu_1 w_1 + \ldots + \mu_s w_s$. Hence, $u_1, \ldots, u_r, w_1, \ldots, w_s$ is a basis of V.

Now suppose $u_1, \ldots, u_r, w_1, \ldots, w_s$ is a basis of V. Let $v \in V$. Then there exist unique $\lambda_1, \ldots, \lambda_r, \mu_1, \ldots, \mu_s \in F$ such that $v = \lambda_1 u_1 + \ldots + \lambda_r u_r + \mu_1 w_1 + \ldots + \mu_s w_s$. Since $u_1, \ldots, u_r$ is a basis of U, there exists a unique $u \in U$ such that $u = \lambda_1 u_1 + \ldots + \lambda_r u_r$. Since $w_1, \ldots, w_s$ is a basis of W, there exists a unique $w \in W$ such that $w = \mu_1 w_1 + \ldots + \mu_s w_s$. Hence, there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. So $V = U \oplus W$.

Therefore, $V = U \oplus W$ if and only if $u_1, \ldots, u_r, w_1, \ldots, w_s$ is a basis of V. $\square$

Exercise 2.6

We use induction on r. Suppose $r = 1$. Then $V = U_1$, so $\dim V = \dim U_1$. Let $k \in \mathbb{N}$. Suppose $V = U_1 \oplus \ldots \oplus U_k$ implies $\dim V = \dim U_1 + \ldots + \dim U_k$. Consider $r = k+1$. Then $V = U_1 \oplus \ldots U_k \oplus U_{k+1}$. Let $W = U_1 \oplus \ldots \oplus U_k$. By the induction hypothesis, $\dim W = \dim U_1 + \ldots + \dim U_k$. Since $V = W \oplus U_{k+1}$, by Exercise 2.4, $\dim V = \dim W + \dim U_{k+1}$. Hence, $\dim V = \dim U_1 + \ldots + \dim U_k + \dim U_{k+1}$. By induction, $V = U_1 \oplus \ldots \oplus U_r$ implies $\dim V = \dim U_1 + \ldots + \dim U_r$ for any $r \in \mathbb{N}$. $\square$

Exercise 2.7

Consider $V = \mathbb{R}^2$, $\vartheta : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $(v_1, v_2)\vartheta = (v_1, 0)$ and $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $(v_1, v_2)\phi = (v_2, 0)$. Note that

$$\begin{aligned}(v_1, v_2)\vartheta^2 &= (v_1, 0)\vartheta \\ &= (v_1, 0) \\ &= (v_1, v_2)\vartheta,\end{aligned}$$

so $\vartheta^2 = \vartheta$. Hence, ϑ is a projection. By Proposition 2.32, $V = \operatorname{Im}\vartheta \oplus \operatorname{Ker}\vartheta$. Note that $(v_1, v_2)\phi = (0,0)$ if and only if $v_2 = 0$. So $\operatorname{Ker}\phi = \{(v, 0) \mid v \in \mathbb{R}\}$. Also, we have $\operatorname{Im}\phi = \{(v, 0) \mid v \in \mathbb{R}\} = \operatorname{Ker}\phi$. Since $\operatorname{Im}\phi + \operatorname{Ker}\phi = \{(v, 0) \mid v \in \mathbb{R}\}$, we have $(0,1) \in V$ but $(0,1) \notin \operatorname{Im}\phi + \operatorname{Ker}\phi$, so $V \neq \operatorname{Im}\phi + \operatorname{Ker}\phi$, and thus $V \neq \operatorname{Im}\phi \oplus \operatorname{Ker}\phi$. $\square$

Chapter 3 Group representations

Exercise 3.1

Suppose ρ is a representation of G over $\mathbb{C}$. Then

$$\begin{aligned} A^m &= (a\rho)^m \\ &= a^m\rho \\ &= 1\rho \\ &= I. \end{aligned}$$

Now suppose $A^m = I$. Let $a^r, a^s \in G$, where $0 \leq r, s \leq m-1$. If $r + s \leq m-1$, then

$$\begin{aligned} (a^r a^s)\rho &= a^{r+s}\rho \\ &= A^{r+s} \\ &= A^r A^s \\ &= (a^r)\rho(a^s)\rho. \end{aligned}$$

If $m \leq r + s \leq 2m - 2$, then

$$\begin{aligned} (a^r a^s)\rho &= a^{r+s}\rho \\ &= a^{r+s-m}\rho \\ &= A^{r+s-m} && \text{(note that } 0 \leq r+s-m \leq m-2 \leq m-1\text{)} \\ &= A^r A^s A^{-m} \\ &= A^r A^s (A^m)^{-1} \\ &= (a^r)\rho(a^s)\rho I^{-1} \\ &= (a^r)\rho(a^s)\rho. \end{aligned}$$

Therefore, ρ is a homomorphism, and hence a representation. □

Exercise 3.2

Since $A^3 = B^3 = C^3 = I$, by Exercise 3.1, ρ_1, ρ_2, ρ_3 are representations of G over $\mathbb{C}$. By computing the powers of the matrices, we have $\operatorname{Ker}\rho_1 = G$, $\operatorname{Ker}\rho_2 = \{a^0\} = \{1\}$ and $\operatorname{Ker}\rho_3 = \{a^0\} = \{1\}$. Therefore, ρ_2 and ρ_3 are faithful, but ρ_1 is not. □

Exercise 3.4

Let $g \in G$. Note that $g\rho = I^{-1}(g\rho)I$ and I is invertible. So ρ is equivalent to ρ, proving (1).

Suppose ρ is equivalent to σ, so we have $g\sigma = T^{-1}(g\rho)T$ for some invertible matrix T. Then $g\rho = (T^{-1})^{-1}(g\sigma)T^{-1}$ and T^{-1} is invertible, so σ is equivalent to ρ, proving (2).

Now suppose ρ is equivalent to σ and σ is equivalent to τ. We have $g\sigma = T^{-1}(g\rho)T$ and $g\tau = S^{-1}(g\sigma)S$ for some invertible matrices T and S. Then $g\tau = S^{-1}T^{-1}(g\rho)TS = (TS)^{-1}(g\rho)TS$ and TS is invertible, so ρ is equivalent to τ, proving (3). □

Exercise 3.6

Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Since $A^4 = B^2 = I$ and $B^{-1}AB = A^{-1}$, by Example 1.4, $\rho : D_8 \to \mathrm{GL}(3, \mathbb{R})$ defined by $(a^i b^j)\rho = A^i B^j$ is a homomorphism. So ρ is a representation of D_8 of degree 3.

We can use the following Python code to compute the kernel of ρ.

```python
import sympy as sp

A = sp.Matrix([
    [0, 1, 0],
    [-1, 0, 0],
    [0, 0, 1]
])

B = sp.Matrix([
    [1, 0, 0],
    [0, -1, 0],
    [0, 0, 1]
])

print("Elements in the kernel ((i, j) means a^i b^j):")
for i in range(4):
    for j in range(2):
        if (A**i) * (B**j) == sp.eye(3):
            print(f"({i}, {j})")
```

We get $\mathrm{Ker}(\rho) = \{a^0 b^0\} = \{1\}$. Therefore, ρ is faithful. □

Exercise 3.7

By the first isomorphism theorem, $G/\,\mathrm{Ker}(\rho) \cong \mathrm{Im}(\rho) \leq \mathrm{GL}(1, F) \cong F^\times$. Since $F^\times$ is abelian, $G/\,\mathrm{Ker}(\rho)$ is also abelian. □

Exercise 3.8

The statement is false. A counter-example is the trivial representation of a non-abelian group. □

Chapter 4 FG-modules

Exercise 4.1

Note that

$$
\begin{aligned}
v_1 \,\mathrm{id} &= v_1, & v_2 \,\mathrm{id} &= v_2, & v_3 \,\mathrm{id} &= v_3,\\
v_1 \begin{pmatrix}1 & 2\end{pmatrix} &= v_2, & v_2 \begin{pmatrix}1 & 2\end{pmatrix} &= v_1, & v_3 \begin{pmatrix}1 & 2\end{pmatrix} &= v_3,\\
v_1 \begin{pmatrix}2 & 3\end{pmatrix} &= v_1, & v_2 \begin{pmatrix}2 & 3\end{pmatrix} &= v_3, & v_3 \begin{pmatrix}2 & 3\end{pmatrix} &= v_2,\\
v_1 \begin{pmatrix}1 & 3\end{pmatrix} &= v_3, & v_2 \begin{pmatrix}1 & 3\end{pmatrix} &= v_2, & v_3 \begin{pmatrix}1 & 3\end{pmatrix} &= v_1,\\
v_1 \begin{pmatrix}1 & 2 & 3\end{pmatrix} &= v_2, & v_2 \begin{pmatrix}1 & 2 & 3\end{pmatrix} &= v_3, & v_3 \begin{pmatrix}1 & 2 & 3\end{pmatrix} &= v_1,\\
v_1 \begin{pmatrix}1 & 3 & 2\end{pmatrix} &= v_3, & v_2 \begin{pmatrix}1 & 3 & 2\end{pmatrix} &= v_1, & v_3 \begin{pmatrix}1 & 3 & 2\end{pmatrix} &= v_2.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
[\mathrm{id}]_{\mathcal{B}_1} &= \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}, & [\begin{pmatrix}1 & 2\end{pmatrix}]_{\mathcal{B}_1} &= \begin{pmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 1\end{pmatrix},\\
[\begin{pmatrix}2 & 3\end{pmatrix}]_{\mathcal{B}_1} &= \begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{pmatrix}, & [\begin{pmatrix}1 & 3\end{pmatrix}]_{\mathcal{B}_1} &= \begin{pmatrix}0 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 0\end{pmatrix},\\
[\begin{pmatrix}1 & 2 & 3\end{pmatrix}]_{\mathcal{B}_1} &= \begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{pmatrix}, & [\begin{pmatrix}1 & 3 & 2\end{pmatrix}]_{\mathcal{B}_1} &= \begin{pmatrix}0 & 0 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\end{pmatrix}.
\end{aligned}
$$

Let $u_1 = v_1 + v_2 + v_3$, $u_2 = v_1 - v_2$ and $u_3 = v_1 - v_3$. Let T be the change of basis matrix from $\mathcal{B}_1$ to $\mathcal{B}_2$. Then

$$
T = \begin{pmatrix}1 & 1 & 1\\ 1 & -1 & 0\\ 1 & 0 & -1\end{pmatrix}
$$

and $[g]_{\mathcal{B}_1} = T^{-1}[g]_{\mathcal{B}_2}T$. So $[g]_{\mathcal{B}_2} = T[g]_{\mathcal{B}_1}T^{-1}$. We can use the following Python code to compute $[g]_{\mathcal{B}_2}$ for each $g \in G$.

```python
import sympy as sp

matrices_B1 = [
    sp.Matrix([
        [1, 0, 0],
        [0, 1, 0],
        [0, 0, 1]
    ]),
    sp.Matrix([
        [0, 1, 0],
        [1, 0, 0],
        [0, 0, 1]
    ]),
    sp.Matrix([
        [1, 0, 0],
        [0, 0, 1],
        [0, 1, 0]
```

```
    ]),
    sp.Matrix([
        [0, 0, 1],
        [0, 1, 0],
        [1, 0, 0]
    ]),
    sp.Matrix([
        [0, 1, 0],
        [0, 0, 1],
        [1, 0, 0]
    ]),
    sp.Matrix([
        [0, 0, 1],
        [1, 0, 0],
        [0, 1, 0]
    ])
]

T = sp.Matrix([
    [1, 1, 1],
    [1, -1, 0],
    [1, 0, -1]
])

matrices_B2 = [ T * A * T.inv() for A in matrices_B1 ]

for A in matrices_B2:
    sp.pprint(A)
    print()
```

We obtain

$$[\mathrm{id}]_{\mathcal{B}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad [(1 \quad 2)]_{\mathcal{B}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & -1 & 1 \end{pmatrix},$$

$$[(2 \quad 3)]_{\mathcal{B}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad [(1 \quad 3)]_{\mathcal{B}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix},$$

$$[(1 \quad 2 \quad 3)]_{\mathcal{B}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad [(1 \quad 3 \quad 2)]_{\mathcal{B}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}.$$

Notice that the first row of $[g]_{\mathcal{B}_2}$ is always $\begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$. This implies $u_1 g = u_1$ for any $g \in G$. $\square$

Exercise 4.2

Let $\lambda \in F$, $u, v \in V$ and $g, h \in G$. Note that $-v \in V$, so $vg \in V$. So axiom (1) holds. Since 1 is an even permutation, we have $v1 = v$, so axiom (3) also holds.

To verify axiom (4), we first assume g is even. Then we have

$$(\lambda v)g = \lambda v = \lambda(vg).$$

Next, we assume g is odd. Then we have

$$(\lambda v)g = -(\lambda v) = \lambda(-v) = \lambda(vg).$$

Hence, axiom (4) holds.

Next, we verify axiom (5). We first assume g is even. Thus,

$$(u+v)g = u+v = ug+vg.$$

Now, we assume g is odd. Thus,

$$(u+v)g = -(u+v) = (-u)+(-v) = ug+vg.$$

So axiom (5) holds.

Finally, we verify axiom (2). First, we assume both g and h are even. Then gh is even. So

$$v(gh) = v = vg = (vg)h.$$

Next, we assume g is even and h is odd. Then gh is odd. So

$$v(gh) = -v = -(vg) = (vg)h.$$

Then, we assume g is odd and h is even. Then gh is odd. So

$$v(gh) = -v = vg = (vg)h.$$

Lastly, we assume both g and h are odd. Then gh is even. So

$$v(gh) = v = -(-v) = -(vg) = (vg)h.$$

Hence, axiom (2) holds.

Therefore, V is an FG-module. □